PORTFOLIO ANALYTICS

# How to read this page

Every part of the page in plain words, for a reader new to investing and to the ideas behind it.

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## Start here

This page takes a handful of investments you choose, looks at how their prices moved, and works out how different ways of splitting money between them would have done. This guide says what each part of the page shows, in plain words, for a reader who has not met these ideas before.

Names in bold, like **Tickers** or **Performance**, are written exactly as the page prints them, so you can find them on the screen. Words in the glossary at the end are explained there.

Every figure describes the past: the prices between two dates you pick. Nothing on the page is a forecast or a recommendation, and nothing you do on it is saved anywhere but in your own browser.

## Fitted, then tested

Most of the page asks one question: how should the money have been split between these investments? The answer is a portfolio, a set of weights saying what share of the money goes in each one, and the page scores portfolios with the Sharpe ratio: the return they earned above a safe rate, for each unit of risk they took. Here risk means how much a portfolio's value swung up and down, which the page calls volatility, so the least volatile mix is the one that swung least. A higher Sharpe ratio is better.

Most figures on the page are in-sample. The weights are chosen by looking at a stretch of prices and then scored on that same stretch, which is like setting an exam after reading the answers. An in-sample figure shows what the best split would have been with hindsight, not what anyone could have earned.

An out-of-sample figure is the fair test: choose the weights on earlier prices, then hold them, unchanged, over later days they have not seen. The **Does it hold up?** chapter runs that test,

and the line at the top of the page quotes the result as it was published.

The gap can be large. Take the **Five mega-caps** set, shares in some of the largest US companies. Fitted to all of its prices up to Sep 2026, at a 2.0% safe rate, the portfolio with the highest Sharpe ratio scored 1.107. Held over years it had not seen, it earned 0.659, below the 0.864 that an even split of the money earned. On the **Cross-asset** set, funds that hold stocks, bonds, gold and property, it went the other way: 0.883 out of sample, against 0.704 for the even split.

So read an in-sample figure as a description of the past. The published out-of-sample result is the fairer test of what choosing weights from past prices was worth over those years. It came out both ways, and it is not a promise about the years ahead.

## The first screen

### The masthead

At the top: the title, who made it and a link to this guide. The basket the figures are about sits in the bar just below; the settings section of this guide walks through it.

On a first visit the page opens on a saved example, so it is never blank: the cross-asset set described above.

### The question and the answer

The page opens on Overview, which asks one question in large type: **Can a smarter mix beat an equal split?** Under it, two sentences answer it. The first is about your tickers, with hindsight: what the starting amount became in the mix with the best trade-off between return and risk, chosen on these very prices, against an even split. The second is Mason's published test, without hindsight. When your tickers are exactly one of the three published sets it quotes that set's figures as published; on any other basket it says what the test found in words and offers to run the same test on your basket.

A word with a dotted underline, such as **With hindsight**, opens a short definition when you tap it.

### The two cards

The first card draws what the starting amount became in each of the three mixes and the benchmark: the best-tradeoff mix (the tangency, highest-Sharpe portfolio), the equal split (equal weight) and the lowest-risk mix (GMV, the least volatile), each line named at its end.

The second, **Hindsight flatters the optimizer** on the example, sets each mix's Sharpe ratio with hindsight (a hollow bar) beside the same mix's published figure without it (a solid bar). On a basket that was never published, your own bars stand alone and the five mega-caps follow as the reference, under their own name.

Every card has two folds under it. **Show the numbers** opens the figures behind the picture; **How it is calculated** opens the method. In the second card's numbers sit one sentence on the highest Sharpe ratio your tickers reached, then four Sharpe ratios: **Equal-Weight Sharpe**, the even split, chosen without looking at the prices; **Tangency Sharpe (in-sample)**, with plus or minus one standard error beside it, a rough sense of how far the figure could be off by luck alone (more, since the mix was chosen on these same prices); **GMV Sharpe (in-sample)**; and the benchmark's own, such as **S&P 500 Sharpe**. The small i beside each explains it. Under them, the published sentence is quoted as it appears on masonjbennett.com, with the published figures; **Walk-forward test** opens the same test, replayed, and the last link opens the method note.

## Why, and what to try next

Three short facts say why the answer comes out as it does: how much of the best-tradeoff mix sits in one holding, how far one published weight swung with the years it looked back over, and what the published one-year tests found.

**Try another basket:** loads one of the three published sets with one tap; or type your own tickers in the bar. The last line says who built the page and links its source code, the method note and this guide.

## The what-if panel

At the end of Build a portfolio, **What if one expected return were different?** opens a panel, **What-if on these prices**. Pick an asset, then slide its expected return, the average yearly return the optimizer assumes it earns, **up to two standard errors either side of its average**. At each step the highest-Sharpe mix is solved again and the sentence prints its new weight in that asset. The least volatile mix never moves, because it does not use expected returns at all.

Why do fitted weights swing so much? The optimizer has to estimate, from the prices, each asset's average return, how much it swings and how each pair moves together. The lines under the slider count those estimates, and say how many years of prices it would take to pin one average return down. That is usually far more than the window holds, so each estimate is rough, and a small change in one can move the weights a lot.

## The settings

The settings sit in the basket bar across the top of the page, which stays in view as you scroll: the tickers, the ready-made baskets and the dates, then **Assumptions**, which opens the rest in

a strip under the bar. Most changes apply at once, with no button to press; new tickers apply when you press Enter. Only a new basket, new dates or a new benchmark fetch prices.

At the bar's right end, **Prices to** says which day's closing prices the figures run to. While the saved example shows, the word **example** sits beside the date; fresh prices replace it when they arrive.

On a phone the bar folds into one row at the top of the page that says what is set; tap **Settings** on it to open every setting in a sheet.

## Tickers

A ticker is the short code a market uses for an investment: AAPL for Apple, VTI for a fund that holds the whole US stock market. Each ticker in the basket is a chip on the bar, and its cross takes it out. **+ Add** opens a box holding the whole list: type between 3 and 10 codes, separated by commas, and press Enter. The codes are Yahoo Finance's, so funds and indexes work too.

## Baskets

**Baskets** fills the list with a ready-made basket. The first three, marked **Published**, are the sets whose out-of-sample result is published; here they are recomputed in-sample on the prices loaded, so their figures are not the published ones. **More baskets** opens the rest.

## Date range

The dates on the bar open **Start** and **End**, the window: the stretch of prices every figure is computed from. The end is today unless you pick a day; then **End today instead** puts it back. A new window fetches prices again.

## Risk-free rate

The risk-free rate is what a safe investment paid over the same time; the page uses short-term US Treasury bills. A Sharpe ratio counts only the return above it. By default it is the average yield over your window, from FRED, the Federal Reserve Bank of St. Louis's public data.

**Use today's rate instead** swaps in the latest yield. Typing a rate fixes it at that, and **Use the rate over the window** goes back. The note under the field always says which rate is in use.

## Benchmark

The benchmark is what everything is measured against: a broad market index, or a fund that tracks one. The choices are S&P 500 (Recommended), Nasdaq 100, Dow Jones, Russell 2000, MSCI World (URTH) and Total Market (VTI).

## Short selling

Off, every weight is zero or more: money only goes into an asset. With **Allow short positions** on, a weight may go below zero. That is a short position: you borrow an asset and sell it, betting its price will fall. The money from the sale goes into the other assets, so their weights alone can come to more than the whole; all the weights together still make the whole. The note under the switch gives the limits. Switching solves the mixes again on the prices already loaded.

## Starting amount

The sum every growth chart starts from. It changes the dollar figures only, never a return or a ratio. The growth charts carry the same field, so a change in either place moves both; it is kept in this browser and never travels in a shared link.

## Explanation level

How much the small i marks beside the figures explain. **Plain**, the default, uses everyday words; **Finance** uses the terms a finance course would; **Formula** gives the formula, and also adds a line under each part's opening sentence with the formula behind its figure. The level changes the explanations, never a figure. This guide uses the Plain wording.

## The chapters

The row of five chapters above the question holds the analysis, in reading order: **Overview**, **Performance**, **Diversification**, **Build a portfolio** and **Does it hold up?**. Each chapter after the first shows its parts one under the other, and most parts open on one sentence, in large type, that states what their figures show, then the charts and tables behind it. Every one of them is computed from the window and the tickers in the settings.

Four portfolios recur across the chapters: equal weight (the even split), GMV (the least volatile mix), tangency (the highest-Sharpe mix) and your own, which the page calls **Custom** and you set in **Build a portfolio**. GMV and tangency are chosen on the window's prices, so their figures are in-sample.

## Overview

### Where should I start?

The chapter the page opens on: the question, the answer, two cards and the baskets to try, described above.

Open Overview

## Performance

### How did each investment and each mix grow, and how far did they fall?

It opens on one sentence: where a starting amount ended highest and lowest. Under it, one chart draws what that amount became. Every further analysis is a row below, a title and its finding in one line; a row opens in place to its chart or table, and the rows you open stay open until you reload. The returns' rows come first, then the falls', under How deep the falls went.

#### Returns & Statistics

### How did each investment grow, and how bumpy was the ride?

The chart shows where a starting amount ended in **Each asset** and in the benchmark, drawn as one line each, or in **The mixes**: the three mixes the page builds, your own mix once you have set one in Build a portfolio, and the benchmark. The buttons above the asset lines switch them on and off, and the box after **Growth of $** changes the starting amount; it is remembered in this browser and never put in a link. **Show the numbers** under the chart gives each line's end figures, and **How it is calculated** says how.

### Calendar years

Each calendar year's return for the four portfolios and the benchmark.

### Month by month

**Months of** picks one of the portfolios for a grid of every month, shaded from loss to gain; the same months are a table under **Show the numbers**.

### Summary Statistics

One row per asset, and a last row for the benchmark: its average yearly return; its volatility, how much it swung, as a yearly percentage (not a gain or a loss: the higher it is, the further the value tended to stray from its average, up or down); its worst and best single days; and two measures of shape. Skewness says whether the biggest moves leaned toward gains or toward losses; excess kurtosis above zero means extreme days came more often than a bell curve would predict.

### Return Distribution

Pick an **Asset** to see its daily returns two ways. **Histogram + Normal Fit** counts how often each size of daily move happened, beside the bell curve a normal distribution would draw. The far ends of the chart are its tails, the rare big days up and down; the chart's title says whether they were fatter or thinner than the curve's, and fatter means big days came more often than the curve allows.

**Q-Q Plot** lines the daily returns up from worst to best and sets each one against where it would fall if returns followed the bell curve; the page calls those places quantiles. If the returns matched the curve, every dot would sit on the straight line. Dots that bend away from it at the ends are days more extreme than the curve allows.

## Risk Analysis

How far did things fall, and how long did the falls last?

The second list of rows. The Drawdowns row names the asset that fell furthest from a high, and when; its chart opens on that asset.

### The portfolios in a fall

Pick a portfolio to list its **Deepest falls** in your window: from which high, to which low, how deep, and how long it took to get back. **Named falls in this window** sets every portfolio against the same well-known market falls, so they are compared over the same days.

### Drawdowns

A drawdown is how far an investment stands below its highest price so far. The chart follows one asset, picked above it, day by day; **Worst drawdown by asset** lists each asset's deepest, with its dates.

#### Rolling volatility

Volatility measured over a moving window of recent trading days, so calm and stormy stretches show as low and high lines. The buttons above it set the window's length.

#### Risk-adjusted metrics

Each asset's Sharpe ratio and its Sortino ratio, and the benchmark's. The Sortino ratio is the same idea counting only the swings below the safe rate, since few investors mind the swings up.

#### CAPM beta and alpha

CAPM, the capital asset pricing model, is the textbook idea that part of an asset's return simply follows the market. Beta says how much an asset tended to move when the benchmark moved: above one, more than the benchmark; below one, less. Alpha is the yearly return left over that the benchmark's moves do not explain. Both describe this window, not a skill that will carry on. The chart draws a dashed line at one, the benchmark's own beta.

Open Performance

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## Diversification

*Which of the investments moved together?*

One part: how closely each pair moved.

### Correlation

*Do these investments rise and fall together?*

Correlation runs from minus one (always moving opposite) through zero (unrelated) to plus one (always moving together). Investments that do not move together are what makes spreading money between them worth anything. The chapter opens on the most and the least correlated pair and a shaded grid of every pair; the same figures are under **Show the numbers**. Three rows below it open in place.

#### Rolling pairwise correlation

Pick two assets and a window length to see how their correlation changed over time, rather than one figure for the whole window.

#### Equal weights against each on its own

What spreading money evenly did in your window: how much the equal split swung in a year, set against how much the assets swung on average one by one. The gap is what diversification took away.

### Daily covariance matrix

The raw material the optimizer works from: a grid with a row and a column for each asset, saying how each pair's daily returns moved together. Where an asset meets itself, the figure is how much it swings on its own. The figures are tiny and hard to read by eye. Correlation divides each one by how much the two assets swing, which makes it run from minus one to plus one and easier to read, but it no longer says how big the swings were; the optimizer needs the covariances for exactly that.

Open Diversification

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## Build a portfolio

*Which mix would have been best, and how does your own compare?*

It leads with the frontier and your own mix beside it; the tables of the fitted mixes follow.

### Portfolio Optimization

*Which mixes of these assets did best, with hindsight, and where does your own sit?*

It opens on the highest-Sharpe mix's weights and its in-sample Sharpe ratio, against the even split's. Everything fitted here is chosen on the same prices it is scored on.

Then the efficient frontier. The chart puts volatility across and return up, so a point further left swung less and a point higher earned more. Each point on the curve is the least volatile mix that reaches a given return. GMV sits at its far left, the least volatile of all. A straight line from the risk-free rate just touches the curve at tangency, the highest-Sharpe mix, which is where the name comes from. The single assets, the even split and your own mix, **Custom**, are marked too, and the line under the chart's title says how far your mix sits from the curve.

Beside it is **Your mix (Custom)**: a slider and a box for each asset, each starting at an equal setting. The page divides every setting by their total, so the shares always add up to the whole. That means raising one asset lowers the share of every other, even the ones you leave alone. The point on the frontier and the five figures under the sliders follow as you move one, and **Reset to equal weights** appears once you have. Under **Show the numbers** are your settings after the division, the shares every figure of your mix uses. Your weights travel in the page's link; your mix is a column of the scorecard below and a line on Performance's growth chart.

### Three portfolios

Side by side: **Equal-Weight Portfolio (1/N)** (the name's ending means the money divided evenly by the number of assets), **Global Min Variance** and **Max Sharpe (Tangency)**. Each shows its return, volatility, Sharpe and Sortino ratios, and **Max DD**, short for maximum drawdown: its deepest fall. The two fitted ones say **weights chosen on this window**; **The walk-forward test** opens what such weights earned on later years.

### Portfolio Weights

How much of the money each mix puts in each asset, as a chart and a table. Under them is **If the true means were anywhere their own noise allows.** The means are the assets' average returns; their noise is how far those averages, worked out from a limited stretch of prices, could be off by chance. The page redraws the averages many times within that noise and solves the mix again each time, to show how far its weights could swing. **Redraw** draws a fresh set.

### Scorecard

Every portfolio in one table: return, risk, falls, how the portfolio moved with the benchmark, and how fragile its weights are. The buttons above it add other ways of building a mix, such as **Risk parity**. **Show every row** shows the rows left out at first, and **What each row means** explains each one.

Besides the usual downloads, **Download Excel, with formulas** gives a workbook that rebuilds most of the scorecard as spreadsheet formulas, working from the closing prices. The fitted weights go in as plain numbers, and a few rows are left out; the workbook's first sheet names them.

### Risk Contribution (PRC)

Pick a portfolio to compare each asset's share of its money with its share of the return and its share of the risk. A small weight in a volatile asset can carry a large share of the risk.

### Summary Comparison

The portfolios' main figures together in one table.

Open Build a portfolio

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## Does it hold up?

*Would the best mix have done as well without hindsight?*

Two parts: the walk-forward test first, then how much the answer moves with the window it is fitted on.

### Walk-forward

*What did a fitted mix earn on days it had not seen?*

This is the out-of-sample test. Weights are chosen on earlier prices, held unchanged for a stretch called a hold, then chosen again on everything up to that point and held again, and so on to the end. Only the returns earned during the holds count.

### Walk-forward test

The switch under the heading has two views. **Your basket** runs the test on the tickers in your settings. **As published** replays the published test.

In your basket's view, **Fit on** sets what each choice of weights may look at, **All prior history** or only the most recent stretch, and **Hold for** sets how long each hold lasts. Both count in returns: there is one daily return for each trading day, so a count of returns is a count of trading days. **Add the four other constructions** adds the scorecard's other ways of building a mix.

### In-sample against out-of-sample

For each mix, the Sharpe ratio on the prices it was chosen from beside the one it earned afterwards. The distance between them is what hindsight was worth.

### Hold by hold

Each hold's dates and the Sharpe ratio each mix earned in it. Under them, **Weights of** picks a fitted mix to show the weights it held in each hold.

### The published figures

The published out-of-sample Sharpe ratios, every set by every mix, in one table.

### How the test was run

The rules of the published run: what each choice of weights saw, how long each hold lasted, and the rate used.

### Five mega-caps · Seven sector ETFs · Cross-asset

One section per published set: each hold, and the weights the two fitted mixes held. **Rerun this set on fresh prices** puts that set into your settings, on the published run's dates and rate with short selling off, and runs the test again on prices fetched now, in your basket's view.

## Sensitivity

How much do the chosen weights depend on which years they were chosen from?

It chooses the GMV and tangency weights again on several windows that all end on the last day, from the most recent year back to the whole window, and opens on how far one asset's weight moves between them.

### Estimation Window Sensitivity

One bar per window for each asset's weight; the switch above the chart moves between **GMV** and **Tangency**. Weights that jump from window to window depend heavily on which years were used.

### Portfolio Metrics Across Windows (in-sample)

Each window's mix, scored on that same window's prices.

**Weights Across Windows**

The weights behind the chart, as tables.

**Custom Portfolio Sensitivity**

Tick **Include Custom Portfolio in sensitivity analysis** to score your own mix, held fixed, on each window too.

Open Does it hold up?

## On every tab

### The i marks

A small i beside a figure or a column explains it. Point at it, move to it with the Tab key, or tap it; tap again or press Escape to close it. Its words follow the explanation level.

### Downloads

Every table has **Download CSV** and **Download Excel** under it, for the table as you see it.

### The link in the address bar

Once you change something, the address bar holds a link to exactly this view: the tickers, dates, rate, benchmark, short selling, your custom weights and the chapter. Send it and the other person opens the same analysis, on the latest prices unless you set an end date. The starting amount is never in it.

### The keyboard

Press Ctrl and K together (Command and K on a Mac) to open a small box that sets the explanation level, opens a chapter or a part of one, or opens this guide, as you type its name.

# Glossary

## Words the page uses

**Annualised**
Turned into a yearly figure. The page counts a year as 252 trading days.

**Basket**
The set of tickers in the settings, the investments every figure is about.

**Benchmark**
What everything is measured against: a broad market index, or a fund that tracks one. No weights are chosen for it; it is there to compare with.

**Capital allocation line**
On the efficient frontier chart, the straight line from the risk-free rate through tangency: every split between the safe rate and the tangency mix.

**Construction**
A rule for building a mix from the prices: equal weight, GMV and tangency, and the others the scorecard can add.

**Correlation**
How closely two sets of daily returns move together, from minus one (always opposite) to plus one (always together).

**Covariance**
How two sets of daily returns move together, including how big their swings were. The optimizer works from these, one for every pair.

**Drawdown**
How far an investment or a portfolio stands below its highest value so far.

**Efficient frontier**
The curve of the least volatile mix for each level of return. A mix below it could have had less risk for the same return.

**Equal weight**
The same share of the money in every asset, kept even each day. Nobody looked at the prices to choose it, which makes it the fair yardstick.

**Estimation window**
The stretch of prices the weights are chosen from; the page also calls it a lookback. Does it hold up? changes it to show how much it matters.

**ETF**
An exchange-traded fund: one ticker that holds many investments and is bought and sold like a share. A sector ETF holds the companies of one industry.

**Excess kurtosis**
A measure of how often extreme days happen. Zero for a bell curve; above zero, big moves came more often than a bell curve allows.

**Expected return**
The average yearly return the optimizer assumes an asset earns. Here it is the asset's average over the window, which is only an estimate.

**GMV**
Global minimum variance: the mix whose value swung least over the window. Choosing it uses how the returns swing and move together, never their averages.

**Hindsight**
Knowing how things turned out. Weights chosen on the same prices they are scored on are chosen with hindsight.

**Hold**
In the walk-forward test, a stretch of days over which a mix is held unchanged, before its weights are chosen again.

**In-sample**
Scored on the same prices the weights were chosen from. It describes the past; it is not what anyone could have earned.

**Index**
A list of companies whose prices are followed together as one number, such as the S&P 500. A benchmark is an index, or a fund that tracks one.

**Mega-cap**
One of the very largest companies, measured by the total value of its shares.

**Normal distribution**
The bell curve. The page draws one beside the real returns to show where they differ from it.

**Optimizer**
The calculation that chooses weights to meet a goal, such as the least volatility or the highest Sharpe ratio.

**Out-of-sample**
Scored on days the weights had not seen when they were chosen. The fair test of a fitted mix.

**Portfolio**
A set of weights across the assets: a way of splitting the money. The page also calls one a mix.

**Rebalanced**
Bought and sold back to the target weights. Every portfolio on the page is rebalanced daily.

**Return**
How much an investment gained or lost, as a share of what was put in. A daily return is one day's change.

**Risk contribution**
An asset's share of a portfolio's risk, which can be far from its share of the money.

**Risk-free rate**
What a safe investment paid over the same time: short-term US Treasury bills. A Sharpe ratio counts only the return above it.

**Rolling window**
A fixed number of recent trading days that moves forward one day at a time, so a figure can be followed through time.

**Short position**
A negative weight: an asset sold after borrowing it, which gains when its price falls. Off unless switched on in the settings.

**Skewness**
Whether the biggest moves leaned toward gains (above zero) or toward losses (below zero).

**Standard error**
How far a figure estimated from a limited stretch of data could be off by chance alone. The page prints it as plus or minus.

**Tangency**
The mix with the highest Sharpe ratio over the window, chosen with hindsight. Also called the maximum-Sharpe portfolio.

**Ticker**
The short code a market uses for an investment, such as AAPL for Apple.

**Trading day**
A day the market was open. Weekends and market holidays have no prices.

**Volatility**
How much returns swing about their average, as a yearly figure. The page's usual measure of risk.

**Walk-forward test**
Choosing weights on earlier prices, holding them over later ones, and repeating to the end: an out-of-sample test.

**Weight**
The share of a portfolio's money in one asset. A portfolio's weights add up to the whole.

## What the i marks say

The page explains these figures in the small i mark beside each one. Here are those explanations in their Plain words; the page also has each in Finance and Formula words.

**Alpha**
The yearly return left over after allowing for how it moved with the benchmark. Above zero, it did better over this window than its beta explains; below zero, worse. It describes the past, not a skill that will last.

**Annual return**
How much money this investment made (or lost) per year on average, shown as a percentage.

**Benchmark Sharpe**
How much return the benchmark earned for each unit of risk over this window, on its own. No weights were chosen for it: it is there for comparison.

**Best and worst month**
The single best and worst calendar month over the window.

**Beta**
How much it tended to move when the benchmark moved: at one, in step with it; above one, further; below one, less.

**Calmar ratio**
Yearly growth divided by the worst fall: how much growth came with each point of the deepest loss.

**Compound annual return**
The steady yearly growth rate that turns the starting amount into the ending amount over this window.

**Correlation**
How closely the portfolio's daily moves follow the benchmark's, from −1 (opposite) to +1 (in step).

**Cumulative return**
How much the starting amount grew or shrank in total over the whole window.

**Downside deviation**
How much the returns swing, counting only the days that fell short of the risk-free rate.

**Equal-weight Sharpe**
How much return an even split of the money across every asset earned for each unit of risk over this window, with the split kept even. Nothing was fitted to pick these weights, so it is the yardstick the fitted mixes are measured against.

**Expected shortfall**
The average loss on the worst 5% of days in this window. Value at risk is the smallest of those losses; this is their average.

**Fit window**
All prior history refits on all the days before each hold, so the fit keeps growing. The other choice refits on only the most recent days, as many as the first fit had, so old years drop out.

**GMV Sharpe (in-sample)**
How much return the mix with the smallest day-to-day swings earned for each unit of risk over this window. The mix was picked on these same prices to swing as little as possible, not to score well, and the score is in-sample: it describes the past and promises nothing.

**Hold length**
How many trading days each set of weights is kept before the next refit. Shorter holds mean more refits, each on more recent prices.

**Information ratio**
The extra yearly return over the benchmark, divided by the tracking error: how far the portfolio's day-to-day returns strayed from the benchmark's.

**Longest drawdown**
The longest stretch spent below an earlier high before climbing back to it, or to the last day if it never did.

**Maximum drawdown**
The biggest drop from a high point to a low point — the worst loss you could have experienced.

**Out-of-sample Sharpe ratio**
The risk-vs-reward score earned on days the weights had not seen: each hold is scored with weights chosen before it began. The days of every hold are joined into one run and scored once.

**Parameters estimated**
How many numbers had to be estimated from the price history to choose these weights. Equal weight and a typed mix need none.

**Positive months**
The share of months that ended with a gain.

**R squared**
How much of the portfolio's daily movement the benchmark explains, from 0% to 100%.

**Risk parity**
A mix in which every asset adds the same share of the portfolio's day-to-day swings, so a calmer asset holds more money and a jumpier one less. It ignores average returns, but its weights come from this window's prices, so its figures are in-sample.

**Sharpe ratio**
A score for how much return came with the risk: the yearly return above the safe rate, for each unit of swing. Higher is better. It describes this window, not the years ahead.

**Sharpe ratio and its standard error**
The risk-vs-reward score, with a ± showing how far it could be off just from the luck of this window. A column whose weights were chosen on these same prices, as the line under its name says, could be off by more.

**Sortino ratio**
Like Sharpe, but it counts only the swings below the safe rate, not the swings up. Higher is better.

**Standard error of the out-of-sample Sharpe ratio**
How far the out-of-sample score could be off from the luck of these particular days alone. Two scores closer together than about one of these are well inside the noise; it is a scale for reading a gap, not a test of one.

**Tangency on shrunk means**
The maximum-Sharpe mix after every asset's average return has been pulled part of the way toward one shared figure, which narrows the gaps between them before the weights are chosen. How far they are pulled is worked out from the data; its figures are in-sample.

**Tangency on the last year**
The maximum-Sharpe mix chosen from the last year of prices only, then held at those weights over the whole window. Its figures are in-sample: the weights were picked with hindsight from part of these same prices.

**Tangency Sharpe (in-sample)**
The highest risk-vs-reward score any mix of these assets reached over this window, with the weights picked from the same prices it is scored on: an in-sample figure, not a forecast. The ± beside it is one standard error, how far the figure could miss by chance alone if the weights had been set before the window began. They were set on these same prices, so the true uncertainty is larger.

**Tangency with a 25% cap**
The maximum-Sharpe mix with no short positions and no asset above 25% of the money. Its weights were chosen on this window's prices, so its figures are in-sample.

**The four other constructions**
Adds the four other ways of choosing weights that the scorecard offers, each chosen again before every hold. One the basket or a fit cannot have is named, with the reason.

**Tracking error**
How much the portfolio's day-to-day returns differ from the benchmark's, scaled to a year.

**Up and down capture**
In months the benchmark rose (up) or fell (down), how much of its move the portfolio made, on average.

**Value at risk**
On the worst 5% of days in this window, the portfolio lost at least this much.

**Volatility**
How wildly the investment's value swings up and down. Higher = more unpredictable.

**Weight lost to a one standard error cut**
Lower the largest holding's expected return by one standard error, a change well inside its own noise, and solve again: how much of its weight it loses.

**Weight range across lookbacks**
Solve the portfolio again on the last 1, 2, 3 and 5 years and on the full window: the most any one asset's weight moves between those answers.

**Weight range across redraws**
Draw expected returns anywhere their own noise allows, many times, and solve on each: the range the largest holding's weight covers, 10th to 90th percentile.

## How the figures are computed

Every formula and data source is listed under **Methodology** at the foot of the page, and of this guide. The code is public, linked beside it.

▼ **Methodology**

**Prices**
Daily adjusted closes from Yahoo Finance (adjusted for splits and dividends), for each ticker and the benchmark.

**Cleaning**
A ticker missing more than 5% of the trading days in the range is dropped (the benchmark never is); the rest are kept only on days when every series has a price. The range must span at least 2 years, and at least 3 tickers and 252 days of prices must remain.

**Returns**
Daily simple returns: each day's price over the day before, minus one.

**Annualisation**
Mean daily return × 252; standard deviation × √252.

**Risk-free rate**
The mean 3-month Treasury yield from FRED (series DGS3MO) over the days the prices cover, with the latest yield shown beside it in the settings; or the rate typed there; the example shown before live prices arrive keeps the rate it was saved with; 2% when FRED cannot be reached.

**Sharpe ratio**
(Annual return − risk-free rate) ÷ annual volatility.

**Sortino ratio**
(Annual return − risk-free rate) ÷ downside deviation: the root-mean-square shortfall of daily returns below the daily risk-free rate, taken over every day (days above it count as zero), × √252.

**Skewness and kurtosis**
The bias-corrected sample statistics; kurtosis is excess kurtosis, so a normal distribution scores 0.

**Maximum drawdown**
The largest fall from a running peak in the value of $1 invested on the first day.

**Portfolios over time**
Fixed weights applied to every day's returns: each portfolio is rebalanced daily to its target weights.

**Portfolio volatility**
$\sqrt{(w^T \Sigma w)}$, from the covariance matrix of daily returns, annualised.

**Optimisation**
The minimum-variance (GMV) and maximum-Sharpe (tangency) portfolios are solved exactly, as quadratic programmes (the Goldfarb–Idnani method, via quadprog). Weights sum to 100%; each lies in [0%, 100%], or in [−100%, 100%] with short positions allowed. When no portfolio earns more than the risk-free rate, the page says so.

**Efficient frontier**
The lowest volatility reachable at 80 target returns, from the GMV portfolio's return up to the highest return the bounds allow, each an exact quadratic programme. The capital allocation line runs from the risk-free rate through the tangency portfolio.

**Risk contribution**
Each asset's share of portfolio variance, $w_i(\Sigma w)_i \div w^T \Sigma w$; the shares sum to 100%.

**CAPM beta and alpha**
An ordinary least-squares regression of the asset's daily excess return on the benchmark's; beta is the slope, alpha the intercept × 252.

**Rolling figures**
Volatility and correlation over trailing windows of 30, 60, 90 or 120 trading days (60 to start).

**Estimation windows**
Trailing 1, 2, 3 and 5 years (252 trading days a year) and the full sample, each optimised and scored on its own returns: in-sample, a description of the fit rather than of what the weights went on to earn.

**Custom weights**

Each weight is held to the bounds, then divided by their total. With short positions allowed, a net exposure of 5% or less, or a weight that normalising pushes past ±100%, is refused with a message.

**Origin**
Began as a graduate course project and was improved afterwards.

Returns are daily simple returns from adjusted closing prices, annualised over 252 trading days. Portfolio weights sum to 100%; each lies between 0% and 100%, or between −100% and 100% with short positions allowed. Every figure describes the period shown and is not a forecast.

Source code on GitHub · masonjbennett.com